

ATS Architecture Diagrams

Day 19 – ATS Documentation & Knowledge Transfer

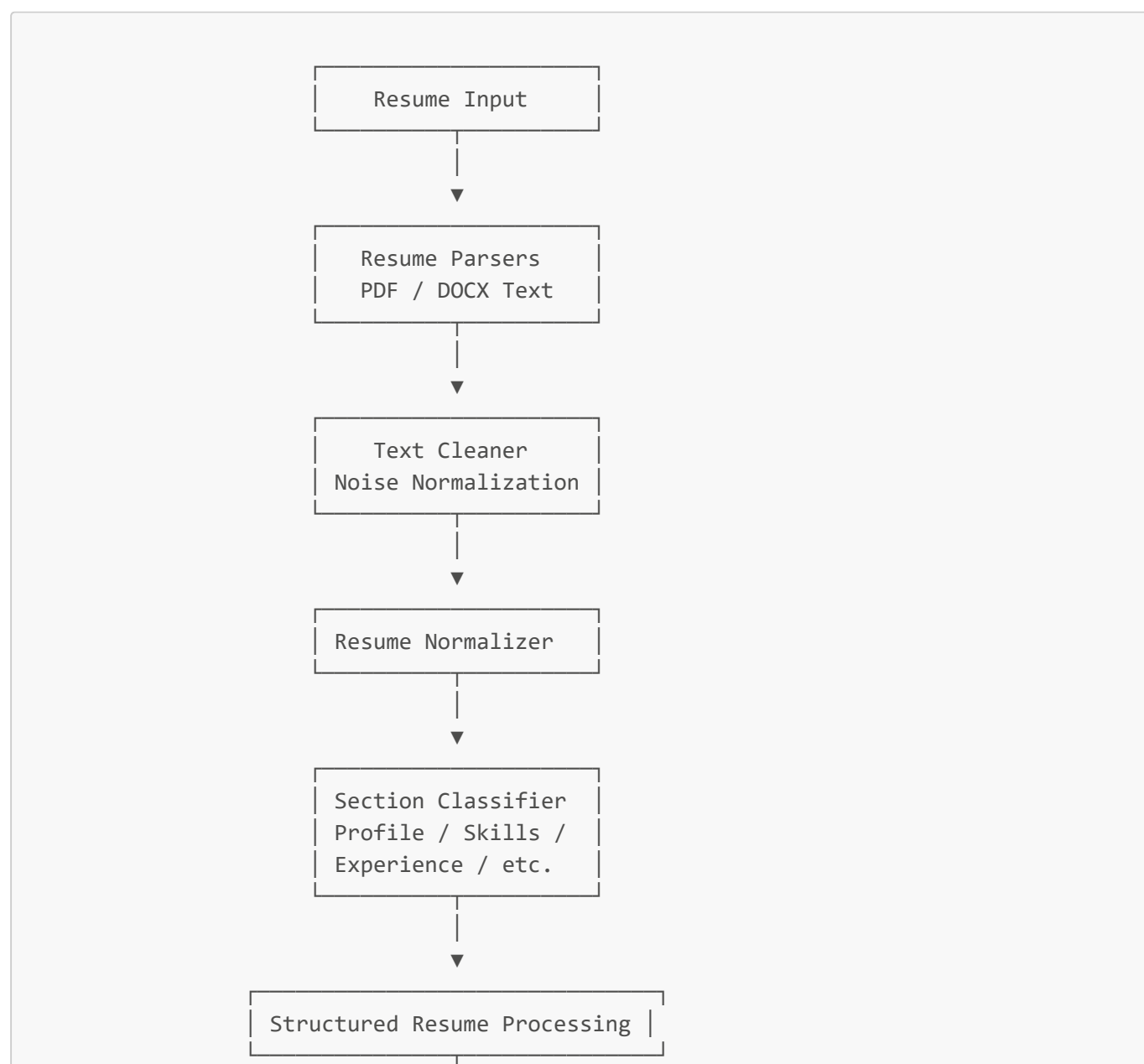
1. Purpose

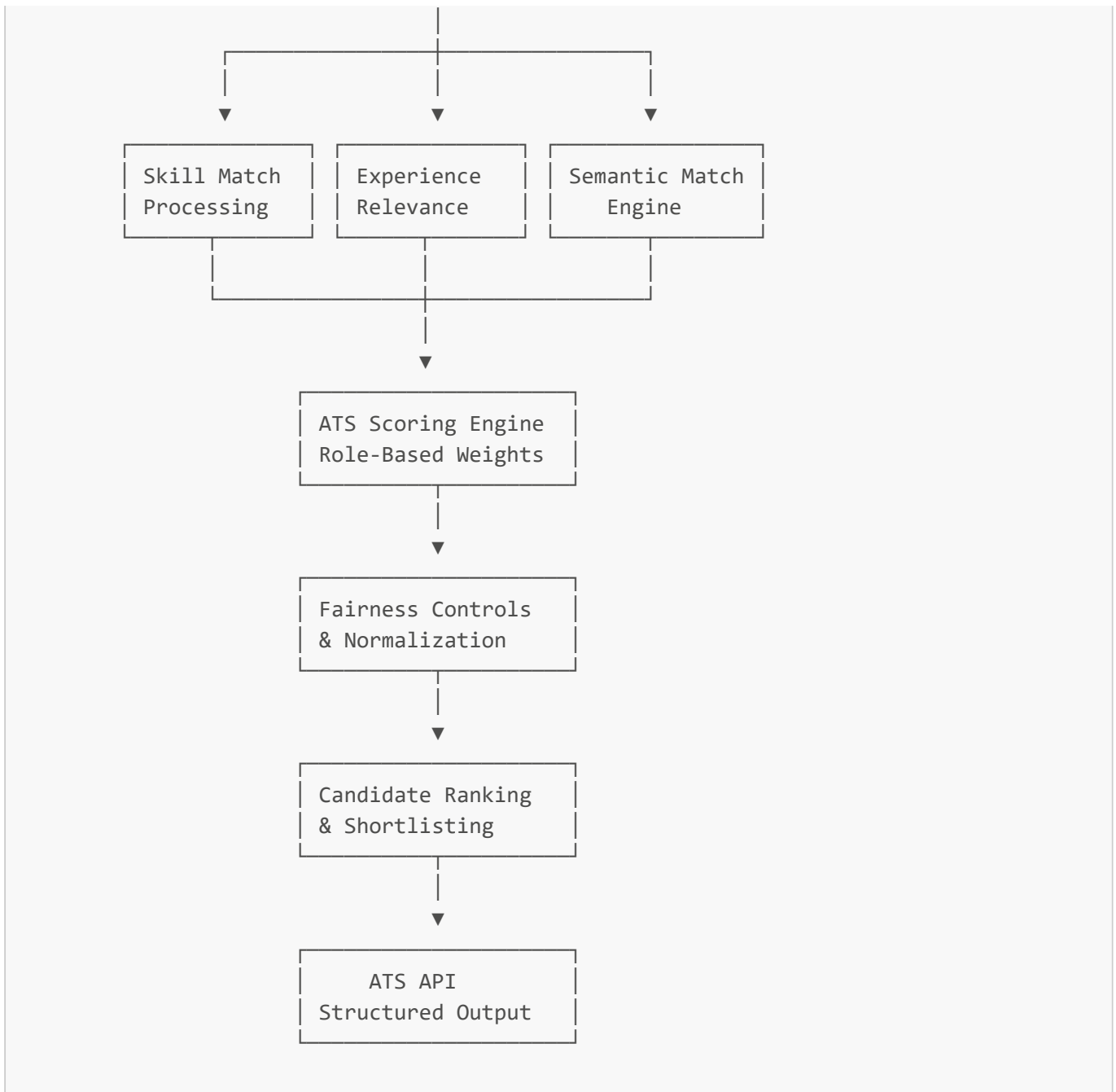
This document provides high-level technical architecture diagrams for the ATS implementation.

The diagrams describe the ATS processing components and their relationships for development and knowledge-transfer purposes.

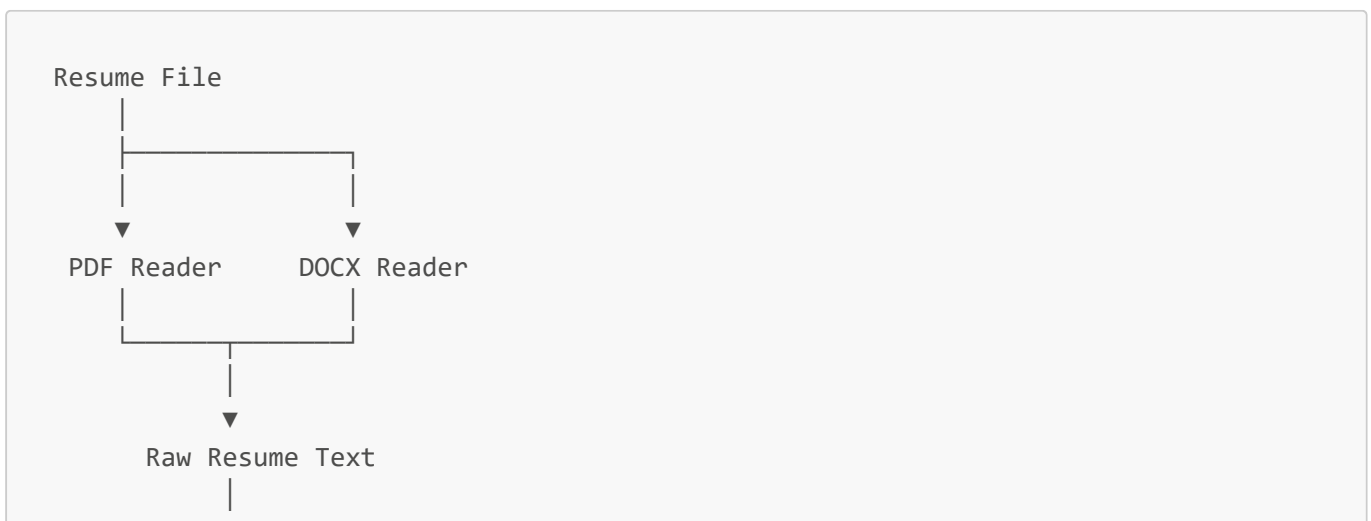
They intentionally focus on the implemented ATS workflow and do not represent confidential organizational infrastructure or undisclosed product architecture.

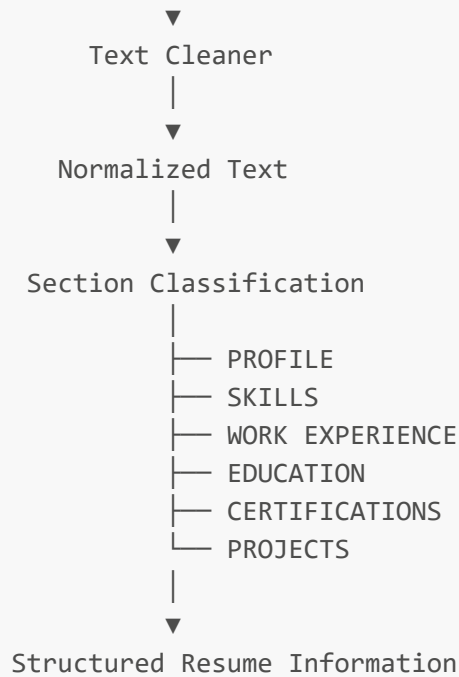
2. High-Level ATS Architecture



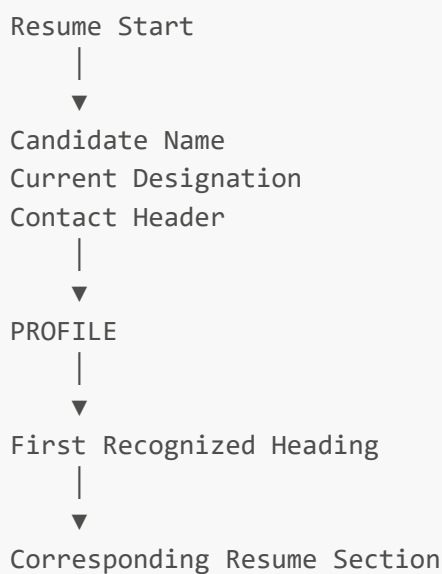


3. Resume Processing Architecture





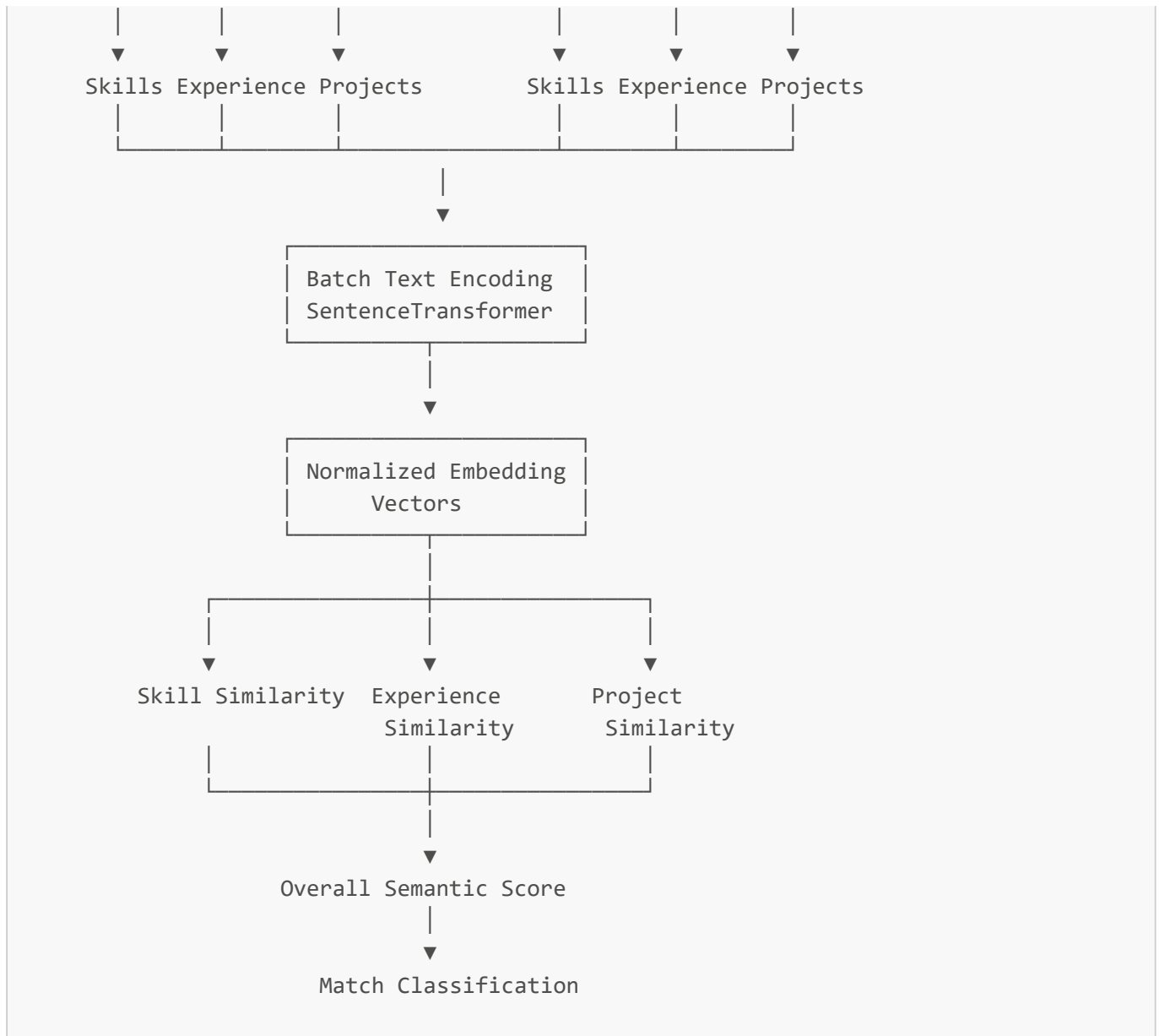
4. Resume Header Processing



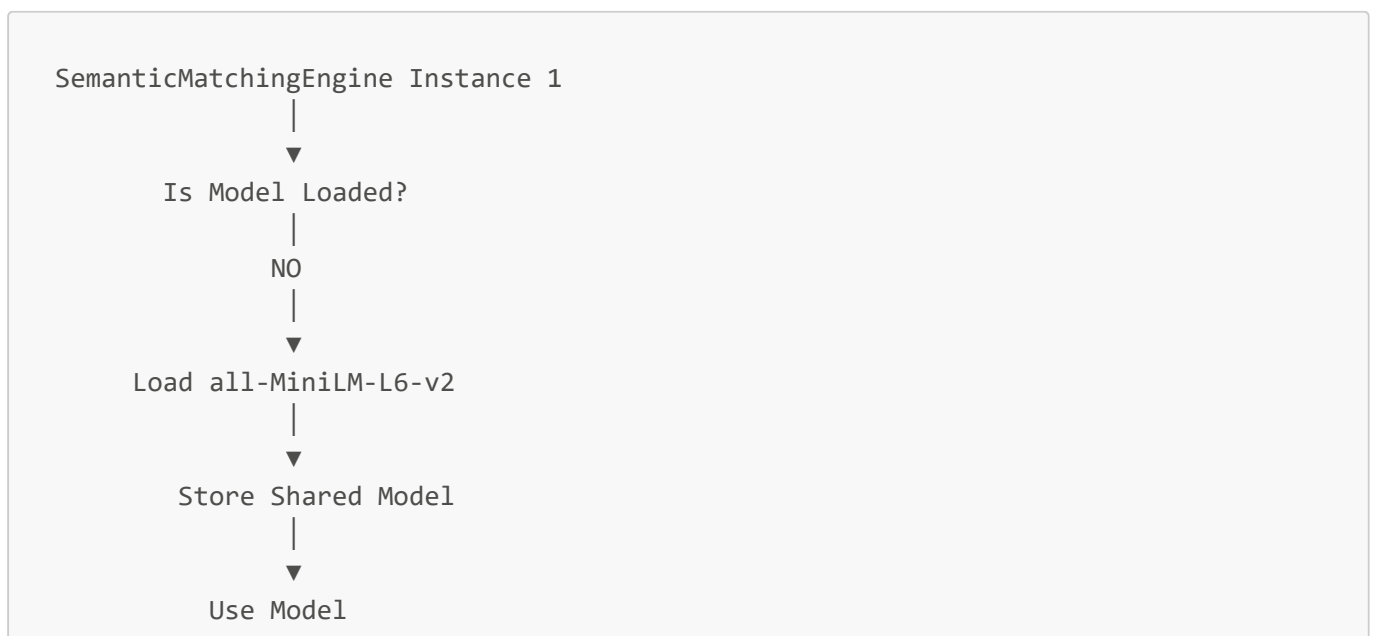
This design prevents resume header information from being unnecessarily classified as UNKNOWN.

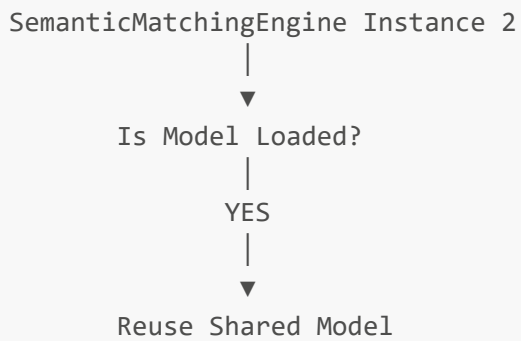
5. Semantic Matching Architecture





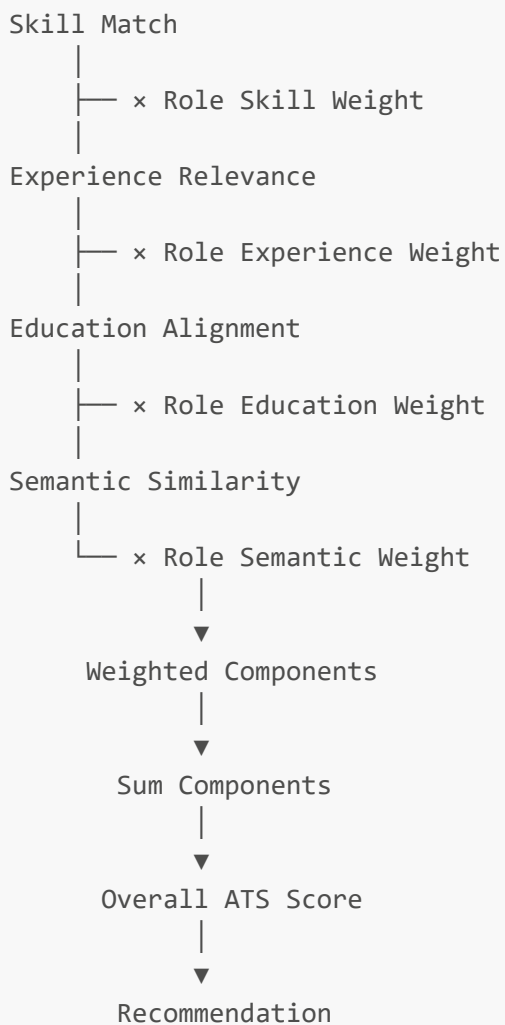
6. Semantic Model Reuse





This avoids unnecessary repeated model initialization within the same running process.

7. ATS Scoring Architecture



8. Scoring Formula

```
Overall ATS Score

=

(Skill Match × Skill Weight)

+

(Experience Relevance × Experience Weight)

+

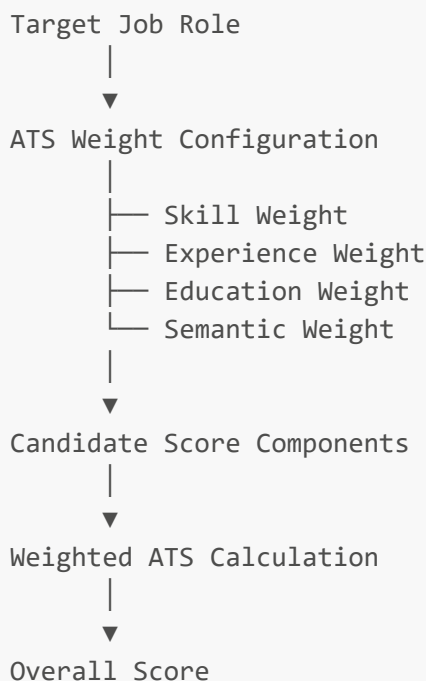
(Education Alignment × Education Weight)

+

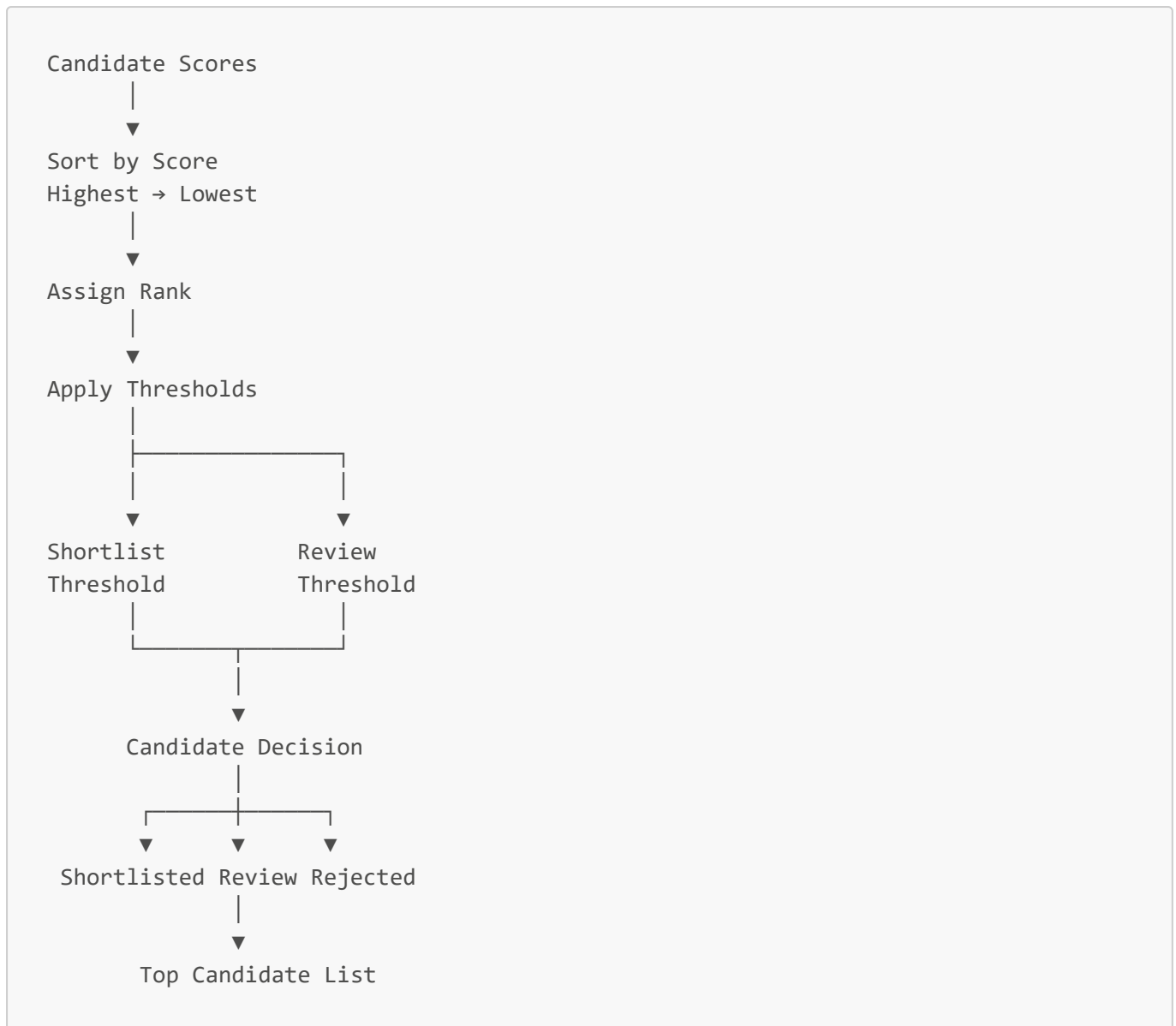
(Semantic Similarity × Semantic Weight)
```

Weights are selected according to the configured target job role.

9. Role Configuration Flow

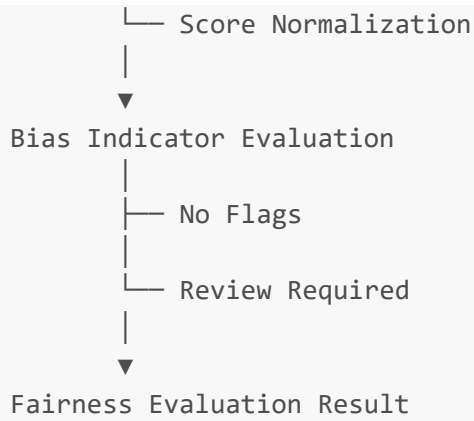


10. Candidate Ranking & Shortlisting Architecture

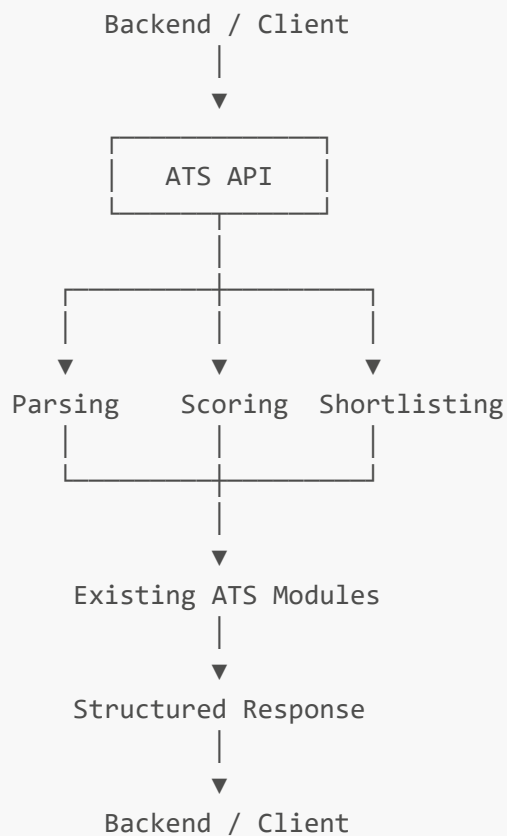


11. Fairness Processing Architecture

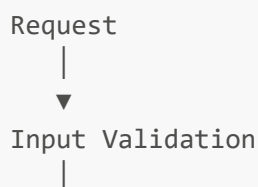


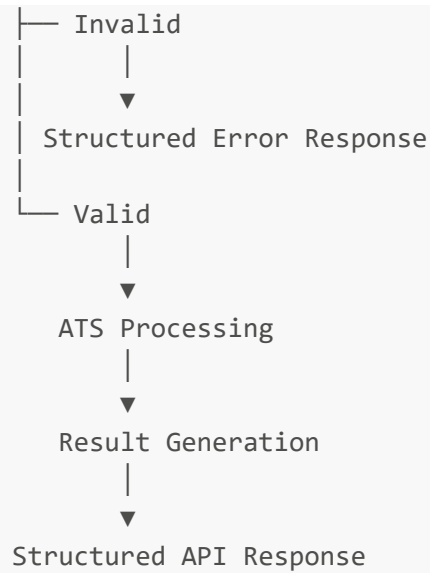


12. ATS API Integration Architecture

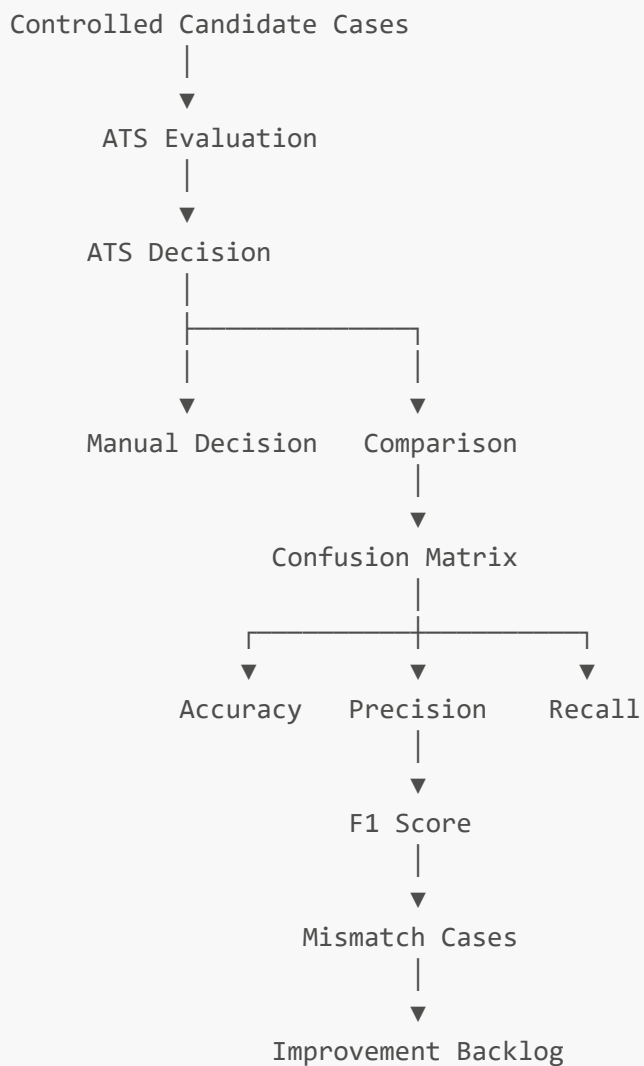


13. API Processing Flow

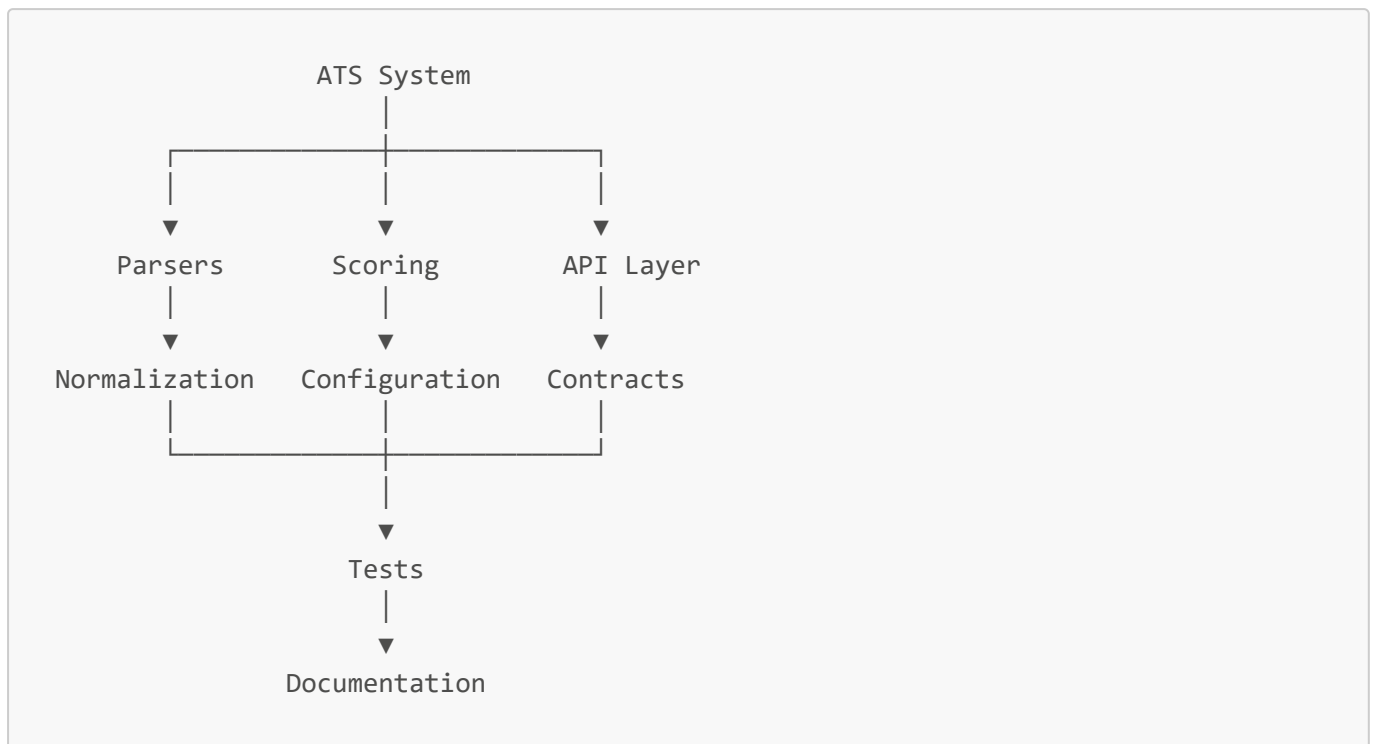




14. ATS Testing Architecture



15. Maintainability View



The modular organization allows individual components to be modified and tested without intentionally redesigning unrelated system areas.

16. Architecture Extension Points

Future development can extend the architecture through:

```
Additional Resume Formats
|
Additional Section Types
|
Improved Entity Extraction
|
Additional Job Roles
|
Role-Specific Thresholds
|
Expanded Fairness Testing
|
Larger Validation Dataset
|
Additional API Capabilities
|
Production Monitoring
```

17. Conclusion

The ATS architecture separates resume processing, semantic analysis, scoring, fairness controls, candidate ranking, shortlisting, API integration, and testing into logical components.

The architecture is designed to remain explainable and configurable while allowing future developers to extend individual processing stages without requiring a complete system rewrite.